eda

March 4, 2026

1 Exploratory Data Analysis (EDA)

This notebook explores the processed dataset from `preprocessing.ipynb`. The goal is to understand feature distributions, spot patterns, and figure out which features might be useful for modelling.

1.1 Workflow

1. Import libraries
2. Load processed data
3. Quick inspection: verify dataset, descriptive stats
4. Univariate analysis: distribution of each feature on its own
5. Bivariate analysis: how each feature relates to `Response`
6. Correlation analysis: heatmap + drill into the strongest pairs

1.2 Step 1: Import required libraries

[1]:
```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

1.3 Step 2: Load the processed training data (Parquet output of preprocessing.ipynb)

[2]:
```
df_train = pd.read_parquet('data/processed/train_final.parquet')
```

1.4 Step 3: Initial inspection of the processed dataset

[3]:
```
# First 10 rows of the processed training set
df_train.head(10)
```

[3]:
```
   id  Gender  Age  Driving_License  Region_Code  Previously_Insured  \
0   0       0   21                1           35                   0
1   1       0   43                1           28                   0
2   2       1   25                1           14                   1
3   3       1   35                1            1                   0
4   4       1   36                1           15                   1
5   5       1   31                1           47                   1
```

```
6   6        0   23                 1            45                    1
7   7        1   47                 1             8                    0
8   8        1   26                 1            28                    1
9   9        1   66                 1            11                    0

   Vehicle_Age  Vehicle_Damage  Annual_Premium  Policy_Sales_Channel  Vintage  \
0            1               1           65101                   124      187
1            2               1           58911                    26      288
2            0               0           38043                   152      254
3            1               1            2630                   156       76
4            1               0           31951                   152      294
5            0               0           28150                   152      197
6            0               0           27128                   152      190
7            1               1           40659                    26      262
8            0               0           31639                   152       36
9            1               1            2630                    26      125

   Response
0         0
1         1
2         0
3         0
4         0
5         0
6         0
7         1
8         0
9         0
```

[4]:
```python
# Descriptive statistics, confirms integer encoding and checks value ranges
df_train.describe()
```

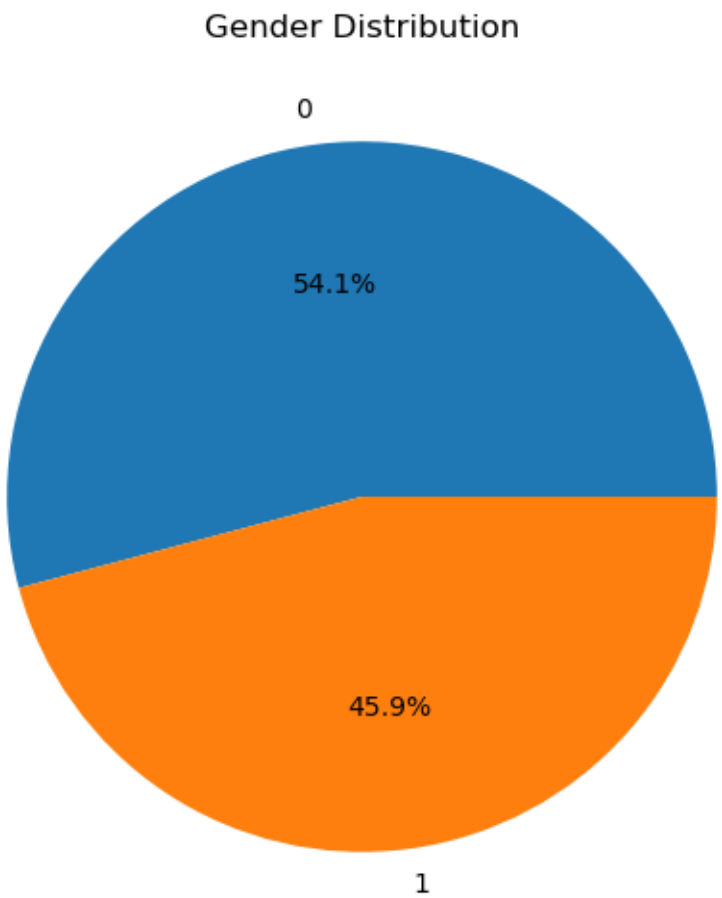
[4]:
```
                 id        Gender           Age  Driving_License  \
count  1.150480e+07  1.150480e+07  1.150480e+07     1.150480e+07
mean   5.752398e+06  4.586490e-01  3.838356e+01     9.980220e-01
std    3.321149e+06  4.982872e-01  1.499346e+01     4.443120e-02
min    0.000000e+00  0.000000e+00  2.000000e+01     0.000000e+00
25%    2.876199e+06  0.000000e+00  2.400000e+01     1.000000e+00
50%    5.752398e+06  0.000000e+00  3.600000e+01     1.000000e+00
75%    8.628598e+06  1.000000e+00  4.900000e+01     1.000000e+00
max    1.150480e+07  1.000000e+00  8.500000e+01     1.000000e+00

        Region_Code  Previously_Insured   Vehicle_Age  Vehicle_Damage  \
count  1.150480e+07        1.150480e+07  1.150480e+07    1.150480e+07
mean   2.641869e+01        4.629966e-01  6.031073e-01    5.026798e-01
std    1.299159e+01        4.986289e-01  5.678559e-01    4.999928e-01
min    0.000000e+00        0.000000e+00  0.000000e+00    0.000000e+00
```

```
25%    1.500000e+01        0.000000e+00  0.000000e+00    0.000000e+00
50%    2.800000e+01        0.000000e+00  1.000000e+00    1.000000e+00
75%    3.500000e+01        1.000000e+00  1.000000e+00    1.000000e+00
max    5.200000e+01        1.000000e+00  2.000000e+00    1.000000e+00

       Annual_Premium  Policy_Sales_Channel       Vintage      Response
count    1.150480e+07          1.150480e+07  1.150480e+07  1.150480e+07
mean     3.046137e+04          1.124254e+02  1.638977e+02  1.229973e-01
std      1.645475e+04          5.403571e+01  7.997953e+01  3.284341e-01
min      2.630000e+03          1.000000e+00  1.000000e+01  0.000000e+00
25%      2.527700e+04          2.900000e+01  9.900000e+01  0.000000e+00
50%      3.182400e+04          1.510000e+02  1.660000e+02  0.000000e+00
75%      3.945100e+04          1.520000e+02  2.320000e+02  0.000000e+00
max      5.401650e+05          1.630000e+02  2.990000e+02  1.000000e+00
```

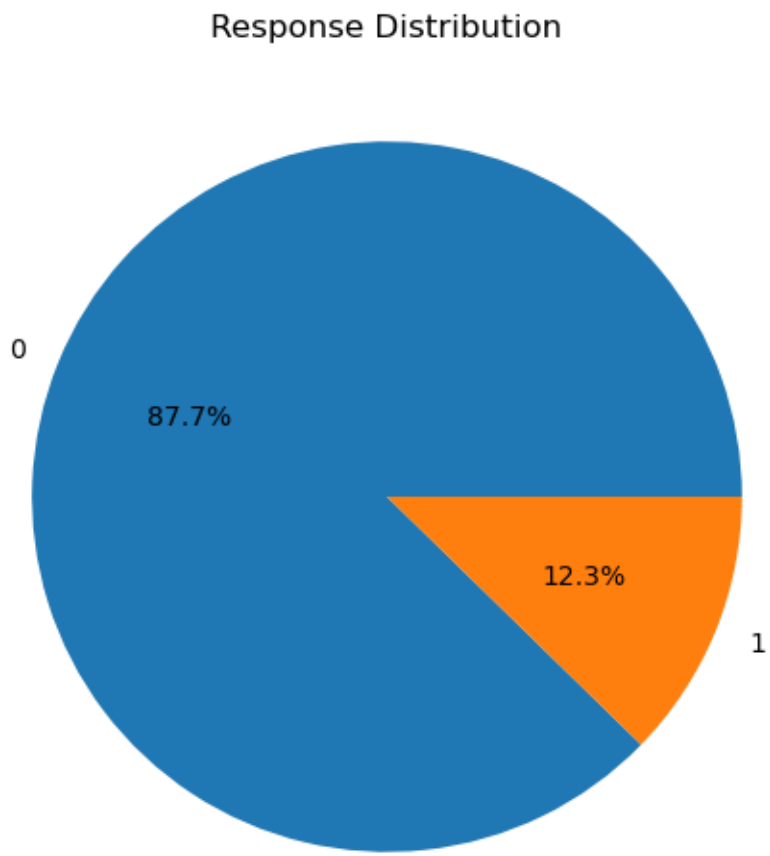
1.5 Step 4: Univariate analysis

Looking at each feature individually to check for skews, imbalances, outliers, etc.

[5]:
```
# Gender distribution, roughly balanced between male and female customers
df_train['Gender'].value_counts().plot(kind='pie', autopct='%1.1f%%',␣
 ↪figsize=(6, 6))
plt.title('Gender Distribution')
plt.ylabel('')
plt.show()
```

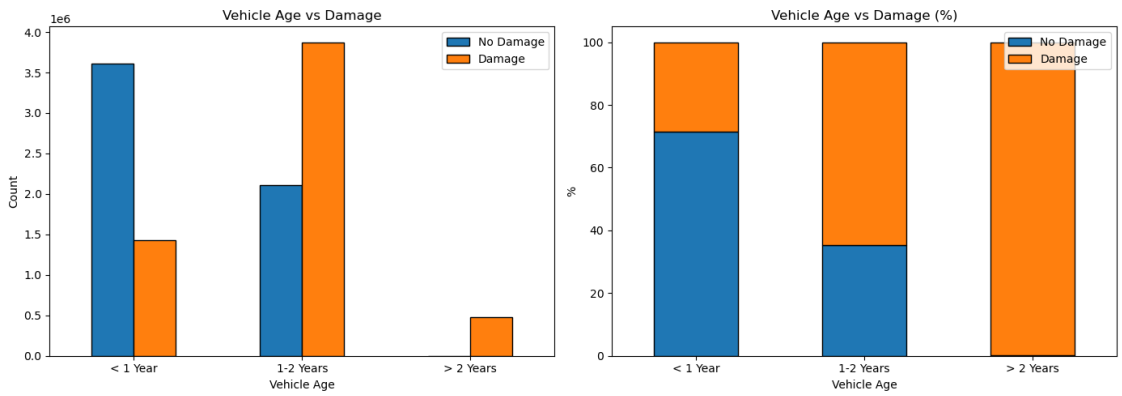


[6]:
```
# Response distribution, confirms class imbalance (~12.3 % positive)
df_train['Response'].value_counts().plot(kind='pie', autopct='%1.1f%%',␣
↪figsize=(6, 6))
plt.title('Response Distribution')
plt.ylabel('')
plt.show()
```

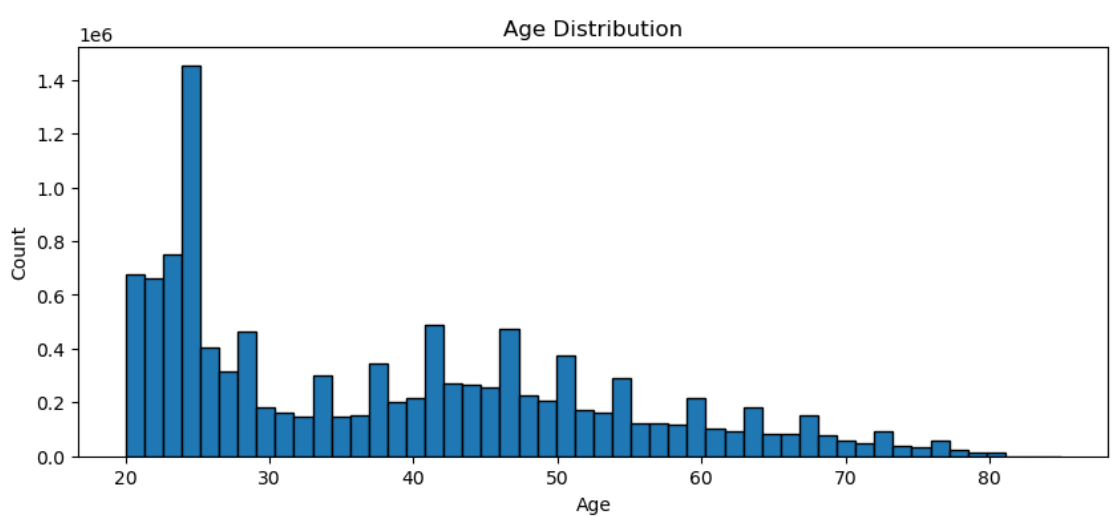


[7]:
```
# Vehicle age vs vehicle damage. Left: absolute counts; right: damage rate by␣
↪vehicle age
fig, axes = plt.subplots(1, 2, figsize=(14, 5))

damage_counts = pd.crosstab(df_train['Vehicle_Age'], df_train['Vehicle_Damage'])
damage_counts.index = ['< 1 Year', '1-2 Years', '> 2 Years']
damage_counts.columns = ['No Damage', 'Damage']

damage_counts.plot(kind='bar', ax=axes[0], edgecolor='black')
axes[0].set_title('Vehicle Age vs Damage')
axes[0].set_xlabel('Vehicle Age')
axes[0].set_ylabel('Count')
axes[0].tick_params(axis='x', rotation=0)

damage_pct = damage_counts.div(damage_counts.sum(axis=1), axis=0) * 100
damage_pct.plot(kind='bar', stacked=True, ax=axes[1], edgecolor='black')
axes[1].set_title('Vehicle Age vs Damage (%)')
```

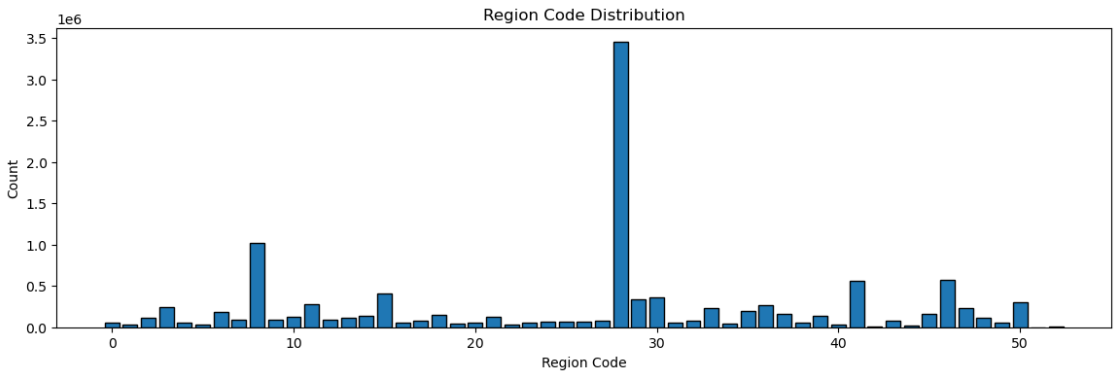
```
axes[1].set_xlabel('Vehicle Age')
axes[1].set_ylabel('%')
axes[1].tick_params(axis='x', rotation=0)

plt.tight_layout()
plt.show()
```



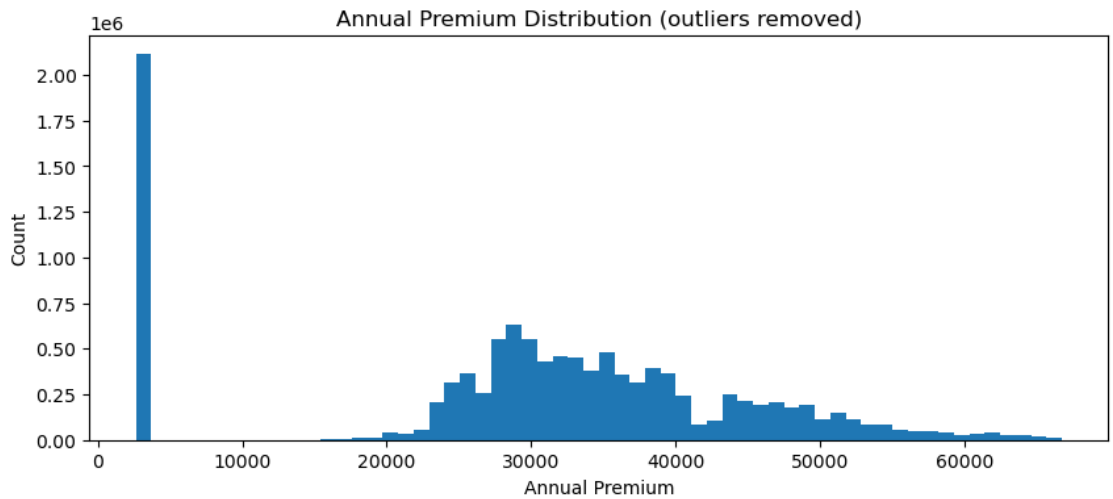
[8]:
```
# Customer age distribution, roughly right-skewed, concentrated in the 20-50␣
↪range
plt.figure(figsize=(10, 4))
plt.hist(df_train['Age'], bins=50, edgecolor='black')
plt.title('Age Distribution')
plt.xlabel('Age')
plt.ylabel('Count')
plt.show()
```



[9]:
```
# Customer count per region, some regions are far more represented than others
region_counts = df_train['Region_Code'].value_counts().sort_index()

plt.figure(figsize=(14, 4))
plt.bar(region_counts.index, region_counts.values, edgecolor='black')
plt.title('Region Code Distribution')
plt.xlabel('Region Code')
plt.ylabel('Count')
plt.show()
```



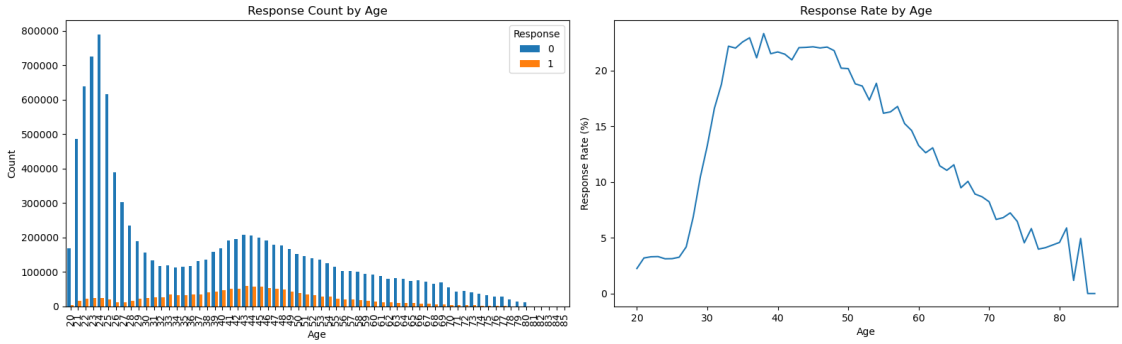
[10]:
```
# remove top 1% outliers before plotting
p99 = df_train['Annual_Premium'].quantile(0.99)
data_clipped = df_train[df_train['Annual_Premium'] <= p99]['Annual_Premium']

plt.figure(figsize=(10, 4))
plt.hist(data_clipped, bins=60, edgecolor='none')
plt.title('Annual Premium Distribution (outliers removed)')
plt.xlabel('Annual Premium')
plt.ylabel('Count')
plt.show()
```



1.6 Step 5: Bivariate analysis

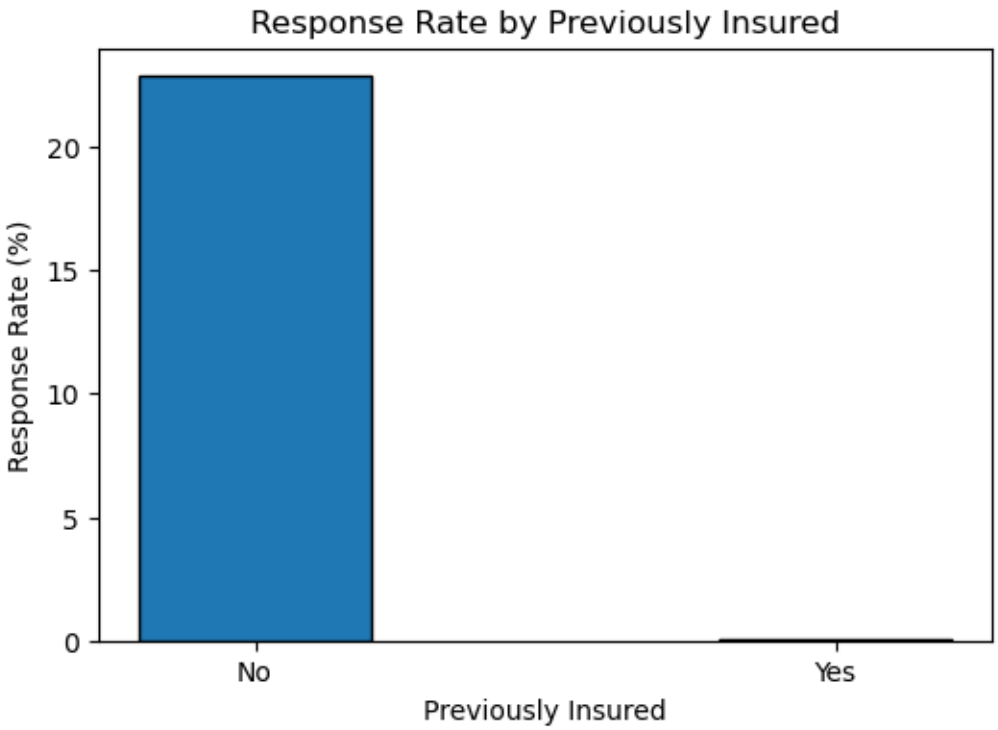
Comparing feature distributions between customers who responded (`Response = 1`) and those who didn't, to see which features are most useful for prediction.

[11]:
```
# Response count and response rate by customer age, shows which age groups are␣
↪most receptive
fig, axes = plt.subplots(1, 2, figsize=(16, 5))

age_response = df_train.groupby('Age')['Response'].value_counts().
↪unstack(fill_value=0)
age_response.plot(kind='bar', ax=axes[0], edgecolor='none', width=0.8)
axes[0].set_title('Response Count by Age')
axes[0].set_xlabel('Age')
axes[0].set_ylabel('Count')
axes[0].tick_params(axis='x', rotation=90)

response_rate = df_train.groupby('Age')['Response'].mean() * 100
axes[1].plot(response_rate.index, response_rate.values)
axes[1].set_title('Response Rate by Age')
axes[1].set_xlabel('Age')
axes[1].set_ylabel('Response Rate (%)')

plt.tight_layout()
plt.show()
```

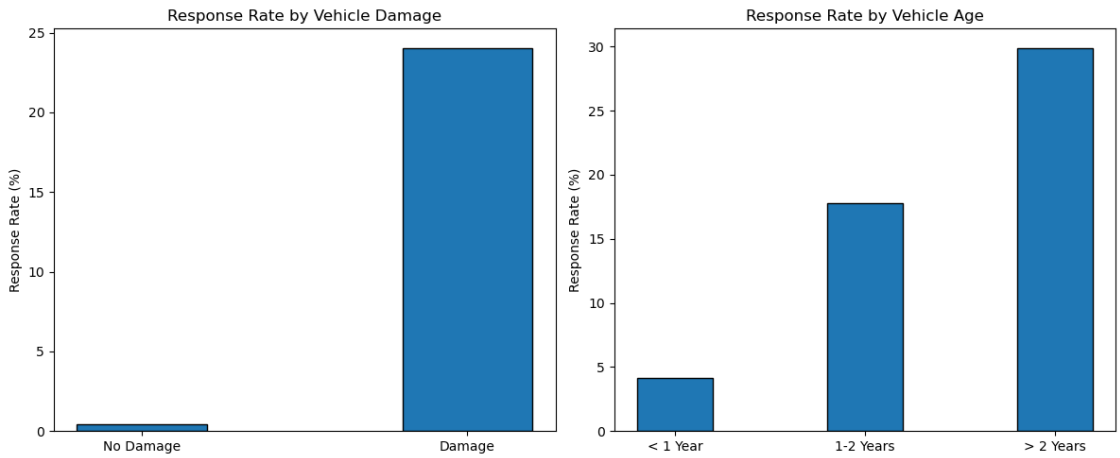


[12]:
```
# Response rate by previously insured status, customers already insured rarely␣
↪respond positively
response_rate_insured = df_train.groupby('Previously_Insured')['Response'].
↪mean() * 100
print(response_rate_insured)

plt.figure(figsize=(6, 4))
plt.bar(['No', 'Yes'], response_rate_insured.values, edgecolor='black', width=0.
↪4)
plt.title('Response Rate by Previously Insured')
plt.xlabel('Previously Insured')
plt.ylabel('Response Rate (%)')
plt.show()
```

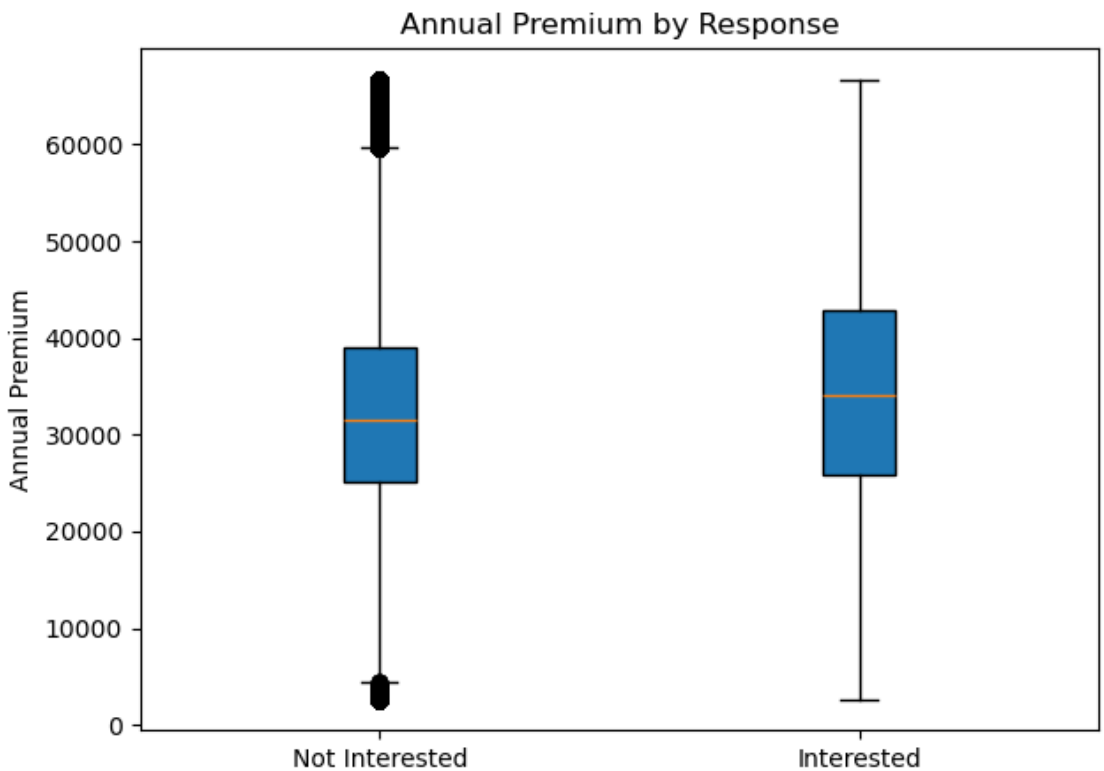
```
Previously_Insured
0    22.849344
1     0.063830
Name: Response, dtype: float64
```



[13]:
```
# Response rate by vehicle damage and vehicle age, damaged / older vehicles␣
↪correlate with higher interest
fig, axes = plt.subplots(1, 2, figsize=(12, 5))

response_by_damage = df_train.groupby('Vehicle_Damage')['Response'].mean() * 100
axes[0].bar(['No Damage', 'Damage'], response_by_damage.values,␣
↪edgecolor='black', width=0.4)
axes[0].set_title('Response Rate by Vehicle Damage')
axes[0].set_ylabel('Response Rate (%)')

response_by_vehicle_age = df_train.groupby('Vehicle_Age')['Response'].mean() *␣
↪100
response_by_vehicle_age.index = ['< 1 Year', '1-2 Years', '> 2 Years']
axes[1].bar(response_by_vehicle_age.index, response_by_vehicle_age.values,␣
↪edgecolor='black', width=0.4)
axes[1].set_title('Response Rate by Vehicle Age')
axes[1].set_ylabel('Response Rate (%)')

plt.tight_layout()
plt.show()
```

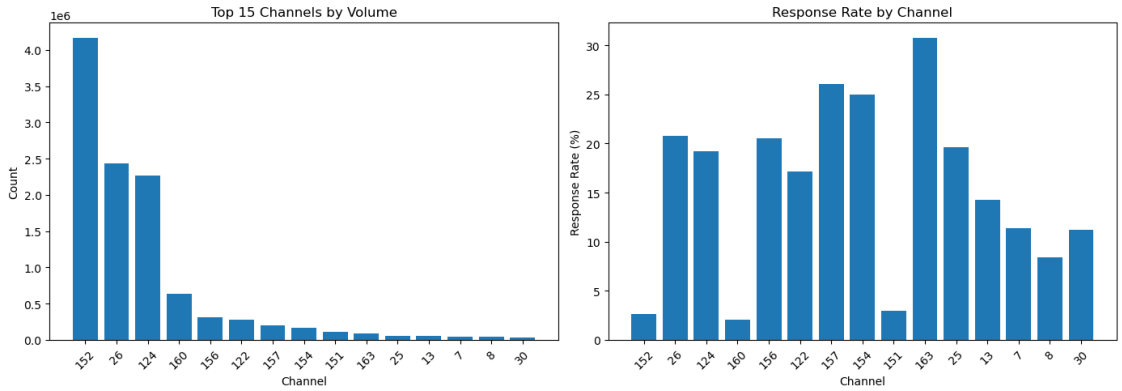


[14]:
```
# Annual premium distribution by response, box plot (top-1 % outliers removed␣
↪for readability)
data_box = df_train[df_train['Annual_Premium'] <= p99]

fig, ax = plt.subplots(figsize=(7, 5))
groups = [data_box[data_box['Response'] == r]['Annual_Premium'].values for r in␣
↪[0, 1]]
ax.boxplot(groups, patch_artist=True)
ax.set_xticks([1, 2])
ax.set_xticklabels(['Not Interested', 'Interested'])
ax.set_title('Annual Premium by Response')
ax.set_ylabel('Annual Premium')
plt.show()
```

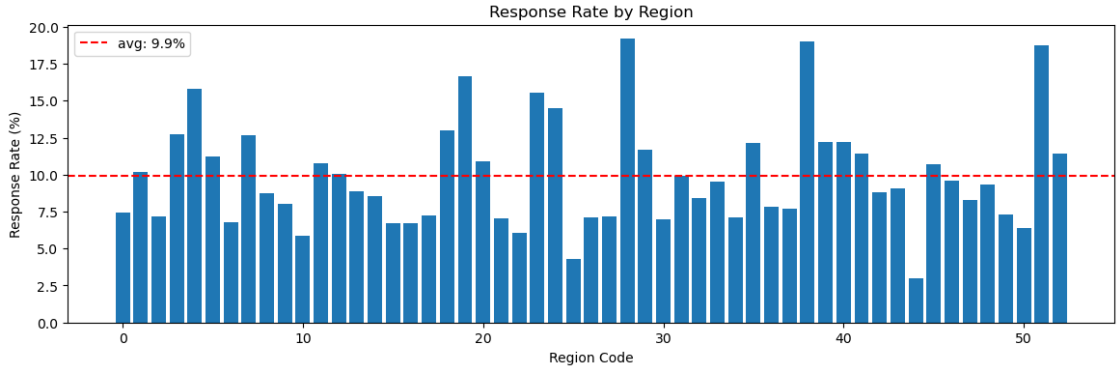


[15]:
```
# Top 15 sales channels by customer volume and their corresponding response␣
↪rates
top_channels = df_train['Policy_Sales_Channel'].value_counts().head(15)

response_by_channel = df_train[df_train['Policy_Sales_Channel'].
↪isin(top_channels.index)] \
    .groupby('Policy_Sales_Channel')['Response'].mean() * 100
response_by_channel = response_by_channel.loc[top_channels.index]

fig, axes = plt.subplots(1, 2, figsize=(14, 5))
axes[0].bar(top_channels.index.astype(str), top_channels.values)
axes[0].set_title('Top 15 Channels by Volume')
axes[0].set_xlabel('Channel')
axes[0].set_ylabel('Count')
axes[0].tick_params(axis='x', rotation=45)

axes[1].bar(response_by_channel.index.astype(str), response_by_channel.values)
axes[1].set_title('Response Rate by Channel')
axes[1].set_xlabel('Channel')
axes[1].set_ylabel('Response Rate (%)')
axes[1].tick_params(axis='x', rotation=45)
```

```
plt.tight_layout()
plt.show()
```



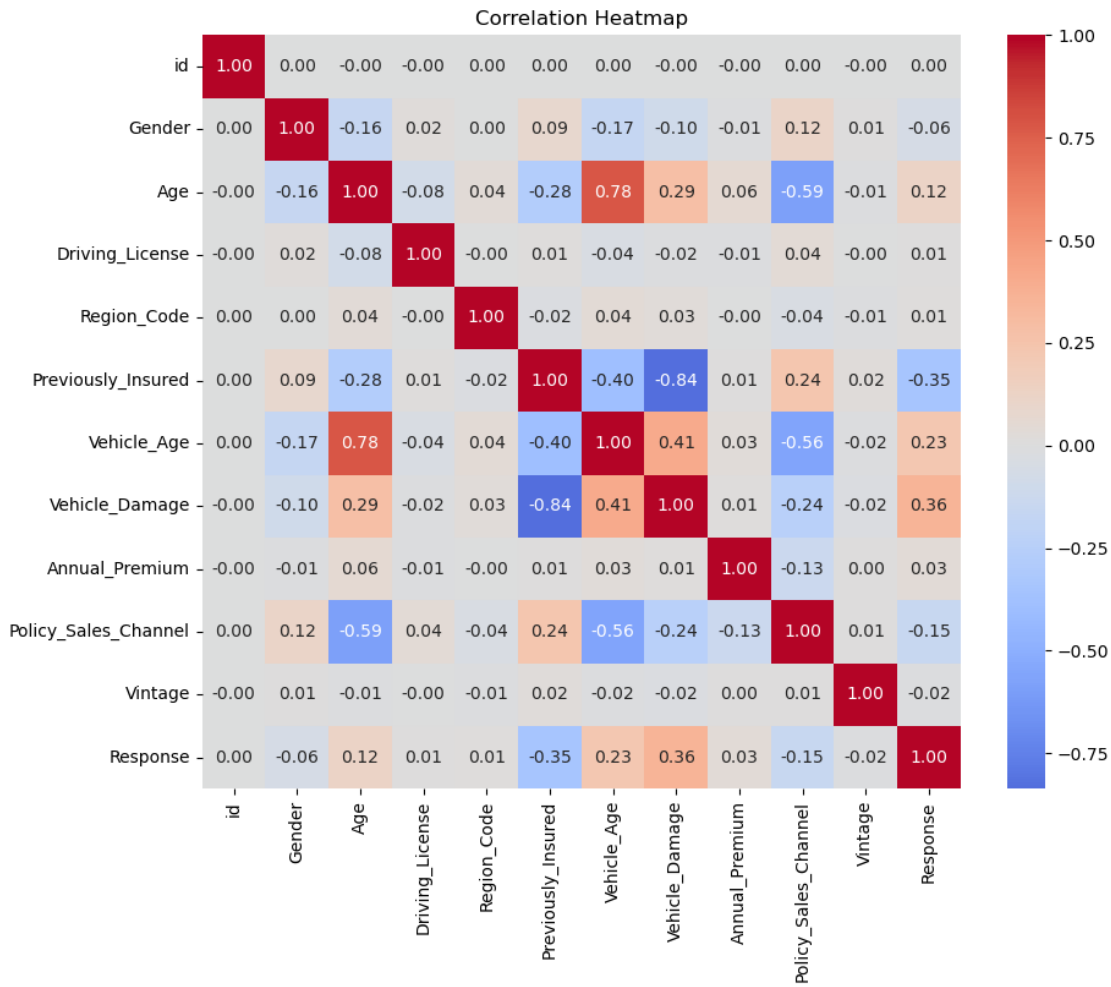
[16]:
```
# Response rate per region, dashed red line marks the overall average (~9.9 %)
response_by_region = df_train.groupby('Region_Code')['Response'].mean() * 100

plt.figure(figsize=(14, 4))
plt.bar(response_by_region.index, response_by_region.values)
plt.axhline(response_by_region.mean(), color='red', linestyle='--', label=f'avg:
 ↪ {response_by_region.mean():.1f}%')
plt.title('Response Rate by Region')
plt.xlabel('Region Code')
plt.ylabel('Response Rate (%)')
plt.legend()
plt.show()
```



1.7 Step 6: Correlation analysis

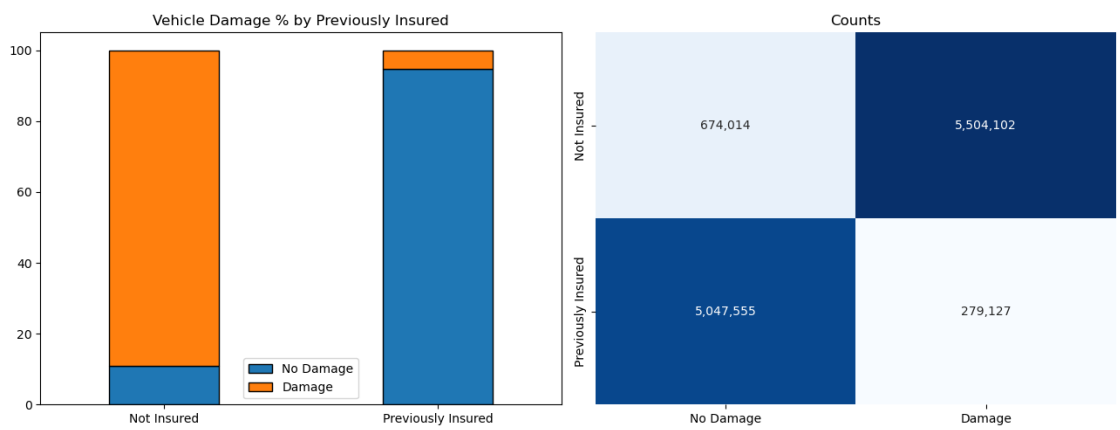
Pearson correlation heatmap, then a closer look at the two strongest pairs: `Previously_Insured` vs `Vehicle_Damage` (r −0.84) and `Age` vs `Vehicle_Age` (r 0.78).

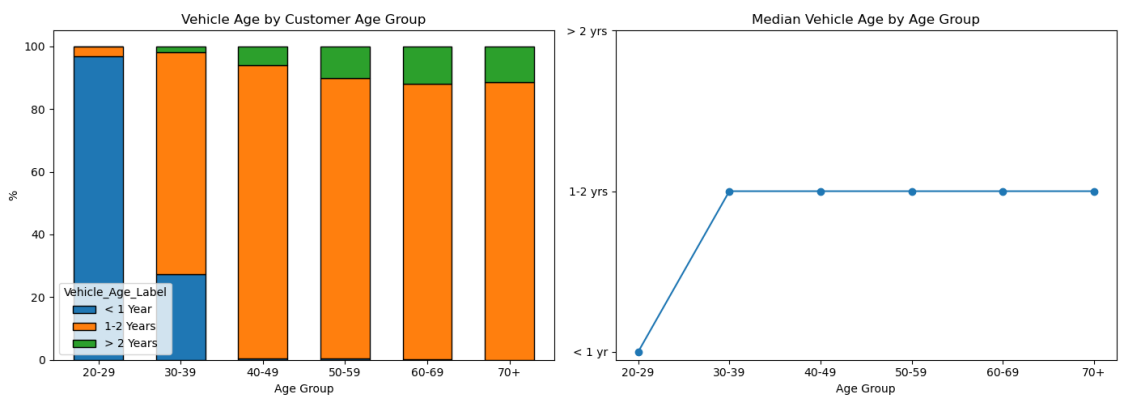
[17]:
```
# Pearson correlation heatmap, highlights strong linear relationships between␣
↪features
corr = df_train.select_dtypes(include='number').corr()

plt.figure(figsize=(10, 8))
sns.heatmap(corr, annot=True, fmt='.2f', cmap='coolwarm', center=0)
plt.title('Correlation Heatmap')
plt.show()
```



[18]:
```
# Previously_Insured vs Vehicle_Damage (r  -0.84)
# Customers who are already insured almost never report vehicle damage and vice␣
↪versa
```

```
fig, axes = plt.subplots(1, 2, figsize=(13, 5))

insured_vs_damage = pd.crosstab(df_train['Previously_Insured'],␣
↪df_train['Vehicle_Damage'], normalize='index') * 100
insured_vs_damage.index = ['Not Insured', 'Previously Insured']
insured_vs_damage.columns = ['No Damage', 'Damage']

insured_vs_damage.plot(kind='bar', stacked=True, ax=axes[0], edgecolor='black',␣
↪width=0.4)
axes[0].set_title('Vehicle Damage % by Previously Insured')
axes[0].tick_params(axis='x', rotation=0)

insured_vs_damage_counts = pd.crosstab(df_train['Previously_Insured'],␣
↪df_train['Vehicle_Damage'])
insured_vs_damage_counts.index = ['Not Insured', 'Previously Insured']
insured_vs_damage_counts.columns = ['No Damage', 'Damage']
sns.heatmap(insured_vs_damage_counts, annot=True, fmt=',', cmap='Blues',␣
↪ax=axes[1], cbar=False)
axes[1].set_title('Counts')

plt.tight_layout()
plt.show()
```



[19]:
```
# Age vs Vehicle_Age (r   0.78)
# Older customers tend to own older vehicles
age_bins = [20, 30, 40, 50, 60, 70, 85]
age_labels = ['20-29', '30-39', '40-49', '50-59', '60-69', '70+']
df_train['Age_Group'] = pd.cut(df_train['Age'], bins=age_bins,␣
↪labels=age_labels, right=False)

vage_map = {0: '< 1 Year', 1: '1-2 Years', 2: '> 2 Years'}
```

```
df_train['Vehicle_Age_Label'] = df_train['Vehicle_Age'].map(vage_map)

age_vs_vehicle_age = pd.crosstab(df_train['Age_Group'],␣
 ↪df_train['Vehicle_Age_Label'], normalize='index') * 100
age_vs_vehicle_age = age_vs_vehicle_age[['< 1 Year', '1-2 Years', '> 2 Years']]

fig, axes = plt.subplots(1, 2, figsize=(14, 5))

age_vs_vehicle_age.plot(kind='bar', stacked=True, ax=axes[0],␣
 ↪edgecolor='black', width=0.6)
axes[0].set_title('Vehicle Age by Customer Age Group')
axes[0].set_xlabel('Age Group')
axes[0].set_ylabel('%')
axes[0].tick_params(axis='x', rotation=0)

median_vage = df_train.groupby('Age_Group', observed=True)['Vehicle_Age'].
 ↪median()
axes[1].plot(median_vage.index, median_vage.values, marker='o')
axes[1].set_title('Median Vehicle Age by Age Group')
axes[1].set_xlabel('Age Group')
axes[1].set_yticks([0, 1, 2])
axes[1].set_yticklabels(['< 1 yr', '1-2 yrs', '> 2 yrs'])

plt.tight_layout()
plt.show()
```